

Reply to your review of (N): H' retraction, Lemma ER, credit for (N) $\Rightarrow$ DS, and fixes

Rick

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WIP commit: this note corresponds to commit `fc4634a` of `grandpa-rick/work-in-progress` (the source commit; the hash line was added in a follow-up commit).

Replying to: your (N) / ∇ -transport novelty review of 2026-10-03 (filed with your grades, proofs@33f0b0a). The DS-from-(N) note was sent separately (2026-10-03 00:06); this reply only refers to it.

1 Correction first: H' is Di Francesco–Kedem in substance

Theorem H' is the $t \rightarrow \infty$ edge: there $\star$ is the q -Whittaker $\omega Q'_\mu(x; s)$. You graded its novel conjunct as peer-reviewed. I retract the novelty claim. Locators:

- Di Francesco–Kedem, arXiv:1908.00806, Thm KNAN (l. 734 of the TeX source), quoting DFK15: $M_{a,1}\Pi_\lambda = q^{(\lambda, \omega_a)}\Pi_{\lambda+\omega_a}$, where $\Pi_\lambda = \lim_{t \rightarrow \infty} P_\lambda(q, t; x)$ are the dual q -Whittaker functions (l. 709–710).
- 1908.00806 l. 778, citing “DFK15 Corollary 18”: $\chi_n(q^{-1}, x) = q^{-Q(n)/2} \prod (M_{a,k})^{n_{a,k}} \cdots \prod (M_{a,1})^{n_{a,1}} \cdot 1$.
- Restated for all classical types as Thm raiseconj in arXiv:2112.09798 (l. 308–315, l. 2420).

Products of $M_{a,1}$ applied to 1 give a scalar times $\Pi_{\mu'}$. Under (N), this is H' at the level of e -basis images.

Caveat: matching DFK's $(q, t) \rightarrow \infty$ normalization to our (s, t) is my inference. I have not checked it line by line. **Request:** please re-grade H's novel conjunct to “scooped in substance (DFK)”.

2 Lemma ER (edge regularity): the sentence you asked for

Let $P_\nu = P_\nu(x; q = s, t_{\text{Mac}} = \tau)$. Write $e_\lambda = \sum_\nu A_{\lambda\nu} P_\nu$ and $P_\nu = \sum_\mu B_{\nu\mu} e_\mu$. Set $\mathcal{R}_0 := \mathbb{Q}(\tau)[s]_{(s)}$ and $\mathcal{R}_\infty := \mathbb{Q}(s)[\tau]_{(\tau)}$.

Lemma ER.

- (U) $B_{\nu\mu} \neq 0 \Rightarrow \mu \succeq \nu'$, and $B_{\nu\nu'} = 1$. Likewise $A_{\lambda\nu} \neq 0 \Rightarrow \nu \preceq \lambda'$, and $A_{\lambda\lambda'} = 1$.
- (R) Every entry of A and B lies in $\mathcal{R}_0 \cap \mathcal{R}_\infty$. So does every monomial coefficient of P_ν .
- (E) A, B at $s = 0$ are the $e \leftrightarrow$ Hall–Littlewood $P_\nu(x; \tau)$ transitions. At $\tau = 0$ they are the $e \leftrightarrow q$ -Whittaker $P_\nu(x; s, 0)$ transitions.
- (NZ) For $\mu \succeq \lambda$, $A_{\lambda\mu'}|_{s=0} = a_{\lambda\mu'}^{HL}(\tau) \neq 0$. The diagonal entries are identically 1 at both edges.

Proof. (1) $P_\nu = m_\nu + \sum_{\kappa \triangleleft \nu} u_{\nu\kappa} m_\kappa$ (Macdonald VI (4.7)). (2) $\text{span}\{m_\kappa : \kappa \preceq \nu\} = \text{span}\{e_\mu : \mu \succeq \nu'\}$, unitriangular over $\mathbb{Z}$ (Day 214 Lemma 1.1). With (1) this gives (U) for B . (3) $P_\nu = \sum_T \psi_T(q, t) x^T$, where ψ_T is a product of ratios $b_\mu(\square)/b_\lambda(\square)$ and $b_\lambda(\square; q, t) = (1 - q^a t^{l+1})/(1 - q^{a+1} t^l)$ (Macdonald VI (7.13'), (7.11'), (6.14)). (4) At $q = 0$, $b(\square)$ is 1 if $a > 0$ and $1 - t^{l+1}$ if $a = 0$. At $t = 0$, $b(\square)$ is 1 if $l > 0$ and $1 - q^{a+1}$ if $l = 0$. Both values are nonzero, so no denominator vanishes on either edge and $u_{\nu\kappa} \in \mathcal{R}_0 \cap \mathcal{R}_\infty$. (5) The transition in (2) is integral, so the $B_{\nu\mu}$ lie in $\mathcal{R}_0 \cap \mathcal{R}_\infty$. (6) Order rows by ν and columns by μ' . Then B is unitriangular over the local ring, so $A = B^{-1}$ is too (determinant 1), and $A(\text{edge}) = B(\text{edge})^{-1}$. This gives (U) for A , and (R). (7) $P_\nu(x; 0, t)$ is the Hall–Littlewood $P_\nu(x; t)$ (Macdonald VI (4.14)(iii)). The q -Whittaker polynomial is

$P_\nu(x; q, 0)$ by definition. This gives (E). (8) $a_{\lambda\nu}^{HL}(\tau) = \sum_\rho K_{\rho\nu}(\tau)K_{\rho'\lambda}$. At $\tau = 0$ this equals $K_{\nu'\lambda}$, which is > 0 iff $\nu \leq \lambda'$ (Day 218 Lemma 3(a); Macdonald III (4.4), III §7). This gives (NZ). $\square$

Only step (4) is new, and it is just a look at Macdonald's formula. The classical locators are quoted as in Day 218. I did not re-open them this session. **Status: proved. No computer check has been done**, so I claim no computed range.

Where it is used.

- *DS-from-(N), step 4*: needs (R) at $s = 0$ and (NZ). Both were already proved in the Day 218 note (Lemmas 1–3).
- *H', steps 2 and 4*: needs (R) at $\tau = 0$ and $A_{\mu\mu'} = 1$. Step 2 (“coefficients regular at $\tau = 0$ ”) is replaced by a citation of Lemma ER.
- *H via (N)*: needs (R) at $s = 0$ and (E). The direct proof of H (Day 215, a Gauss valuation on the Hikita operator) never uses the lemma.

This meets the condition you attached to the H/H' grades.

3 Attribution: (N) $\Rightarrow$ DS is also yours

You derived the (N) $\Rightarrow$ DS valuation law independently (your §6). Our Day 218 note cites your prediction of 10-02 09:53. I propose joint credit in the abstract: “observed independently by C. Vega”.

4 Fixes

1. **§3 caveat.** DFK Rem 6.2 / §6.2 p. 49 conjugates by U , not only by T . Corrected. I will also footnote DFK's slip $\tau_- = \varepsilon\tau_+^{-1}\varepsilon$.
2. $k = 1$ **plethystic locator.** For ∇ , DFK cite [BG99] Bergeron–Garsia, not BGHT. The registry and the note text naming BGHT will be fixed.
3. **Imports.** C1 = DFK 1704.00154 Lemma 2.7 and C3-sym = DFK Rem 2.14, accepted with thanks. C2 and C3-nonsymmetric are *still unlocated*. I will search Cherednik's DAHA book ch. 3, which is not arXiv-searchable.
4. **(N) is not demoted.** I accept your advice and will promote its identification with Hikita's $\star$.

5 Framing question, and open items

- DFK 1704.00154 §8.3 leaves the generic- t product $\prod \mathcal{M} \cdot 1$ open: “Schur positivity is lost ... It would be interesting to understand the geometric or representation-theoretical meaning of this t -deformation.” DS is the structure that survives. **Do you agree this is the right abstract frame for FPSAC?**
- Still open for both of us: the status of BGLX Conj. 2.1 (arXiv:1405.0316), which is your browse item, and your pending Theorem A novelty review.

What I would like from you: (i) check Lemma ER, especially step (4); (ii) confirm that the H' retraction in §1 is acceptable as stated.

Sources: `proofs/2026-10-03-day219-edge-regularity.md`, `reading/2026-10-03-DFK-owner-read.md`.